AIORI Hackathon 3 · Smart Cities

PS-003

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Abstract

Problem Description

Centralised forecasting

Edge cameras and roadside sensors stream raw data to a central cloud. This uses high cellular bandwidth and concentrates private travel logs in one place.

Federated learning

Models train locally on edge devices. Only mathematical weight updates go to a central server, so raw logs stay at each intersection.

This project

A federated simulator over a public traffic dataset, split across 3 to 5 virtual intersections and evaluated with FedAvg.

Goal: compare federated and centralised traffic-flow prediction, and show how non-IID data and communication overhead affect accuracy and convergence.

Context

Background

Prediction should scale without centralising raw traffic data.

1. Intelligent Transportation Systems need timely traffic-flow predictions to optimise traffic control.
2. Streaming everything to one cloud costs bandwidth and creates a single point of failure.
3. Federated learning trains at the edge, but non-IID data across intersections can slow or degrade convergence [2].

Related work

Literature and RFC Review

Work	Contribution	Use in this project
McMahan et al., 2017 [1]	FedAvg: local training with iterative model averaging	Core algorithm of the simulator
Li et al., 2020 [2]	Survey of statistical heterogeneity and communication cost	Frames the IID vs non-IID and bytes-per-round study
Li et al., FedProx [3]	Generalises FedAvg for non-identically distributed clients	Reference point for client drift under non-IID data
Li et al., DCRNN [4]	Data-driven traffic forecasting on the METR-LA benchmark	Public dataset and MAE / RMSE metrics

Approach

Proposed Solution

Data and clients

METR-LA speed readings [4], with sensors grouped into virtual intersections. Chronological split: earlier windows train, later windows test.

Model and training

One compact forecasting model. Each round, clients run local SGD and the server applies FedAvg, weighted by local sample count [1].

Experiments

Three setups: centralised baseline, FedAvg with IID clients, FedAvg with non-IID clients. Every setup runs with 3 random seeds.

207
loop detectors

5 min
reading interval

34,272
time steps

3
random seeds

Before and after

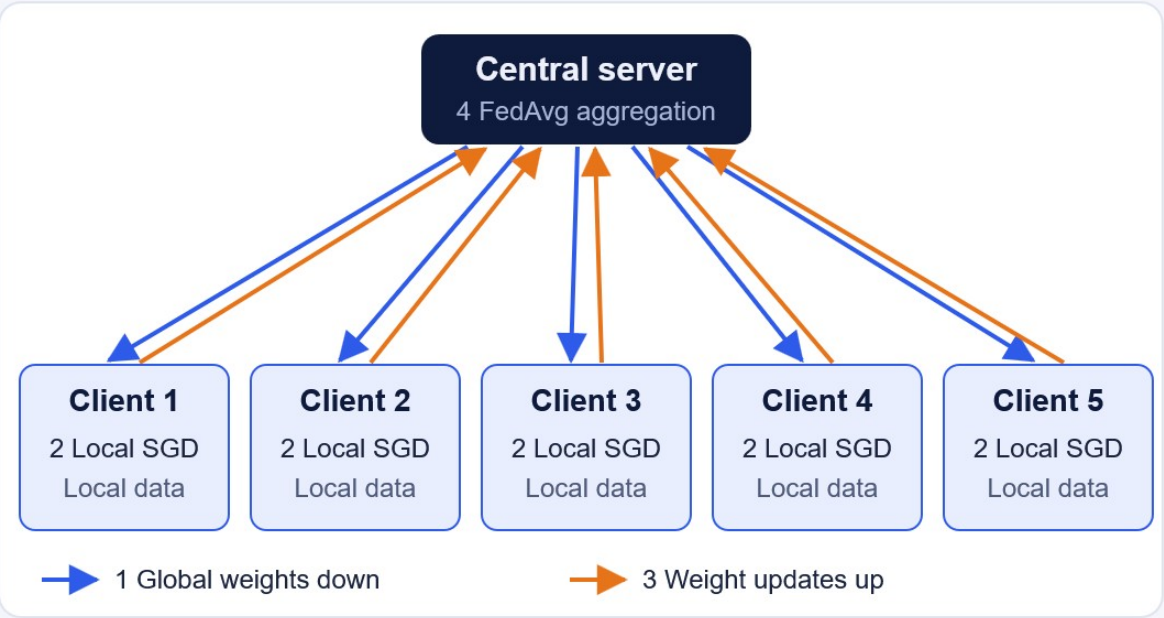
Optimisation: Centralised vs Federated

Feature	Centralised baseline	Federated (IID)	Federated (non-IID)
Where data lives	Central cloud	Edge client nodes	Edge client nodes
Network traffic	Raw data stream (high)	Weight updates only (low)	Weight updates only (low)
Privacy exposure	Travel logs held centrally	Raw logs stay local	Raw logs stay local
Accuracy (MAE, RMSE)	Reference baseline	Compared with baseline	Measured through client drift
Convergence	Not round-based	Rounds to target error	Rounds to target error, compared with IID

The accuracy and convergence rows are what the experiments measure; no results are claimed here.

Architecture design

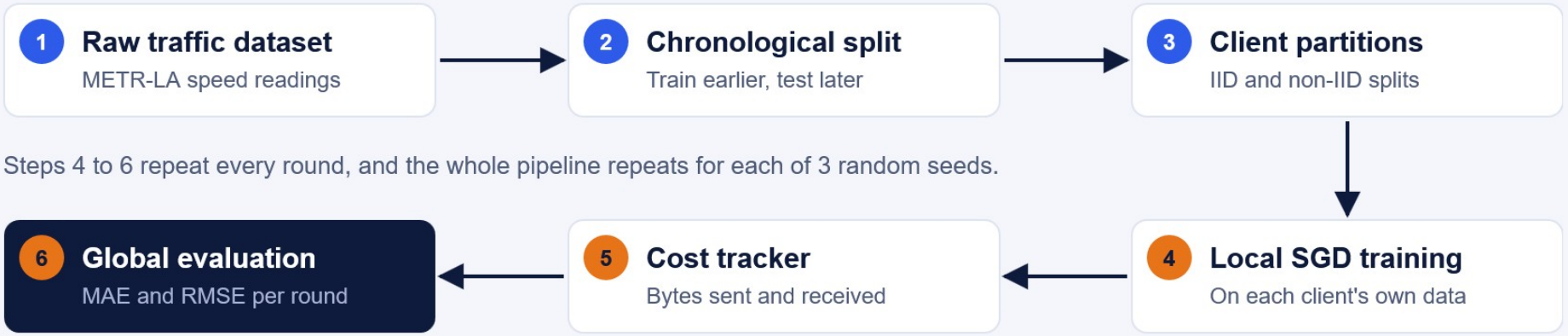
One Federated Round



1. **Broadcast.** The server sends the global weights to every client.
2. **Train locally.** Each client runs SGD on its private traffic logs.
3. **Upload.** Clients return only their updated weights.
4. **Aggregate.** The server applies FedAvg, weighted by local sample count.

Data flow

Data Flow Diagram



Steps 4 to 6 repeat every round, and the whole pipeline repeats for each of 3 random seeds.

Leakage control

Training windows come before test windows, and normalisation uses training data only.

IID vs non-IID

IID: random sensor assignment. Non-IID: grouped by traffic behaviour, so clients differ.

Repeatability

Results are reported as mean and spread across 3 random seeds.

Evaluation

Evaluation Plan and Results

MAE
Mean absolute error of test-period forecasts.

RMSE
Root mean squared error; penalises large misses.

Bytes per round
2 × K × P × 4 bytes for K clients and P float32 parameters.

Convergence
Rounds until test error settles near its final value.

Setup (mean over 3 seeds)	MAE	RMSE	Bytes per round	Rounds to converge
Centralised	pending	pending	raw data upload	not applicable
FedAvg, IID	pending	pending	pending	pending
FedAvg, non-IID	pending	pending	pending	pending

Plan

Deployment Timeline

1

Data preparation

Preprocess the dataset and partition clients, IID and non-IID.

2

Simulator build

Centralised baseline and FedAvg simulator around one compact model.

3

Experiments

Run 3 random seeds and track MAE, RMSE and bytes per round.

4

Submission

Private GitHub repo named after the team, with aiori-hackathon added as collaborator and owner.

Deliverables: federated simulator, centralised and federated baselines, forecasting model, IID vs non-IID evaluation, communication and convergence analysis, and a final report across 3 seeds.

References

Bibliography

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," AISTATS, 2017.

[2] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated Learning: Challenges, Methods, and Future Directions," IEEE Signal Processing Magazine, 37(3), pp. 50-60, 2020.

[3] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," MLSys, 2020.

[4] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting," ICLR, 2018.

Thank You for Your Attention

Questions are welcome.

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